

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

B. Tech. EE Sem-VIII Theory Examination

(April-May, 2018)

EE408 Power System Stability and Control

Date: 04/05/2018, Friday

Time: 01:30 p.m. To 04:30 p.m.

Maximum Marks: 70

Instructions:

1. The question paper comprises two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Figures to the right indicate full marks.
4. Make suitable assumptions and draw neat figures wherever required.

SECTION – I

- Q - 1** Draw the block diagram of generalized control system of the power system and briefly explain the function of each block. Also explain operating states of power system. **8**
- Q – 2 (A)** A 50 MVA, 30-kV, three-phase, 60-Hz synchronous generator has synchronous reactance of 9Ω per phase and a negligible resistance. The Generator delivering 40MW at a terminal voltage of 30kV, Excitation voltage of 23.56kV and the power angle is 17° . Compute the power angle, armature current and power factor when the field current is adjusted for the following excitations. **7**
1. The excitation voltage is decrease to 79.2%.
 2. Find the minimum excitation below which the generator will lose synchronism.

OR

- (A)** Derive electric power (P_e) Vs – Load Angle (δ) equations for cylindrical rotor synchronous machine. **7**
- Q – 2 (B)** A 60 Hz synchronous generator having inertia constant $H=9.94$ MJ/MVA and a transient reactance $X_d'=0.3$ p.u. is connected to an infinite bus through a purely reactive circuit as shown in figure 1. Reactances are marked on the diagram on a common system base. The generator is delivering real power of 0.6 p.u., 0.8 power factor lagging to the infinite bus at a voltage of $V=1$ p.u. Assume the p.u. damping power coefficient is $D=0.138$. Consider a small disturbance of $\Delta\delta=10^\circ=0.1745$ radian. Obtain undamped angular frequency of oscillation, damping ratio, damped frequency of oscillation, response time constant and settling time. **8**

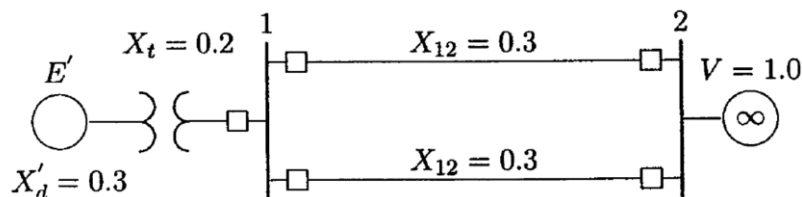


Fig. 1

OR

- (B)** Derive the equation of resultant armature mmf of synchronous generator and draw vector diagram for one phase of a cylindrical rotor generator. **8**

Q – 3 Attempt Any Two 12

- (A) With the use of two bus network, derive the relation between receiving end power and voltage. Also, draw and explain the relationship between power, voltage and current as a function of load impedance.
- (B) Draw and explain the steps to obtain P-V and Q-V curve for the voltage stability analysis of the power system.
- (C) Explain voltage collapse scenario in mature power system and also give the corrective measures to prevent voltage collapse.

SECTION – II

Q – 4 Derive the block diagram of a SMIB system for small signal stability analysis using classical generator. 8

Q – 5 (A) Derive the equations to analyzed transient stability of power system using point-by-point method. List out the necessary assumptions and draw the required curves for rotor speed, rotor angle and accelerating power. 7

OR

(A) Explain Euler's and modified Euler's method for the solution of swing equation. 7

Q – 5 (B) With the help of equal area criteria, explain the case of application of three phase fault at the end of two parallel lines for transient stability analysis. Derive the equation of critical clearing angle for this case. 8

OR

(B) For the system given in figure 2, where a three phase fault is applied at the point P as shown. 8

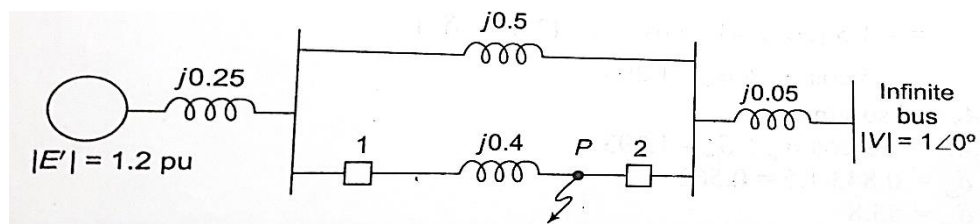


Fig. 2

Find the critical clearing angle for the clearing the fault with simultaneous opening of the breakers 1 and 2. The reactance values of various components are indicated on the diagram. The generator is delivering 1.0 pu power at the instant preceding the fault.

Q – 6 Attempt Any Two 12

- (A) Discuss the special case about the requirement of PSS. Give the structure of PSS and explain the function of each block of PSS.
- (B) Derive the swing equation of synchronous machine.
- (C) Give the detailed classification power system stability and define each term.
